Surgical Continuum Instrument

Department of Mechanical Engineering Summer Research Program 2025

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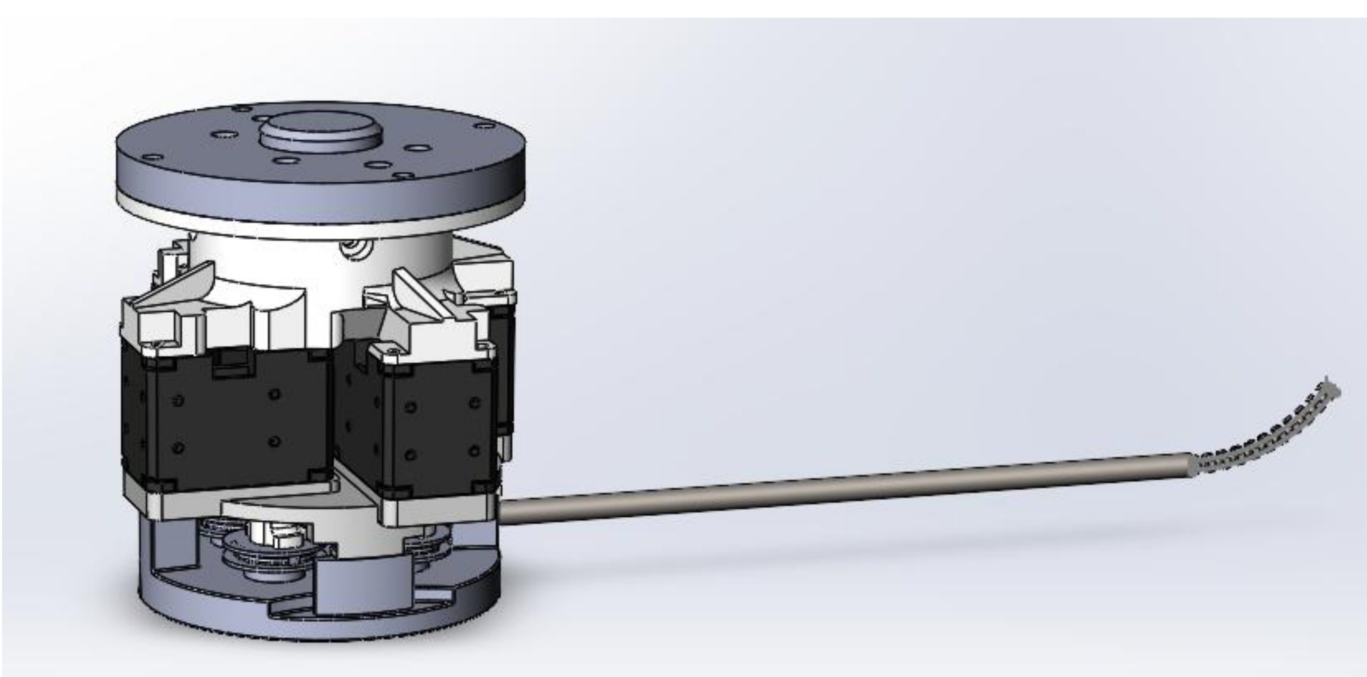
Introductions

Surgical continuum instruments are often used in minimally invasive surgeries (MIS). Compared to conventional open surgeries, MIS requires making smaller incisions during surgeries, and would require shorter recovery period and reduce complications. The instrument studied compose of four pullies controlling the tip of the arm and four motors, each powering an individual pulley.



Goals of my research:

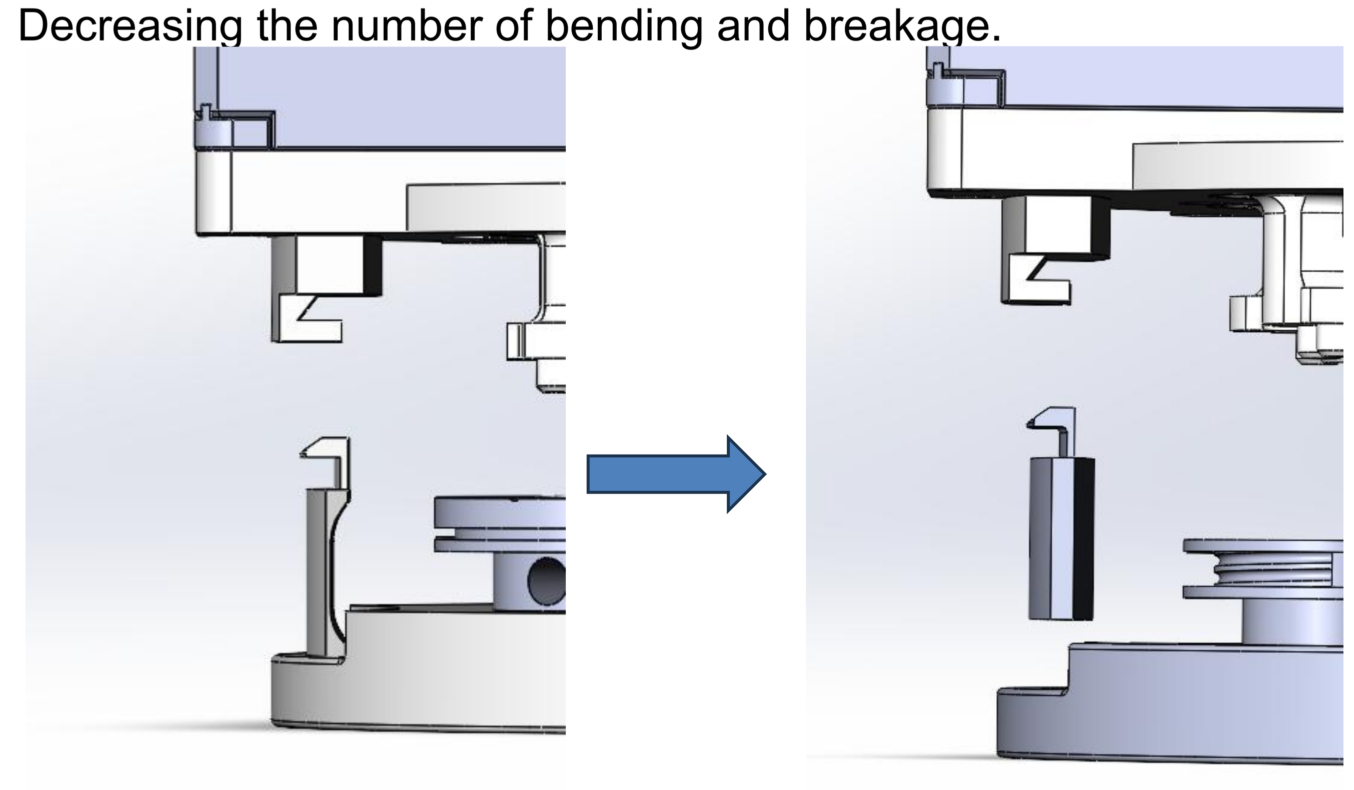
- Improve the mechanics of the overall instrument
- Program the instrument

Mechanics

Attachment Piece

In the original snap fit design, in order to allow for the flexibility of the material, the design was quite thin overall. Leading to snapping and breakage during the attachment process.

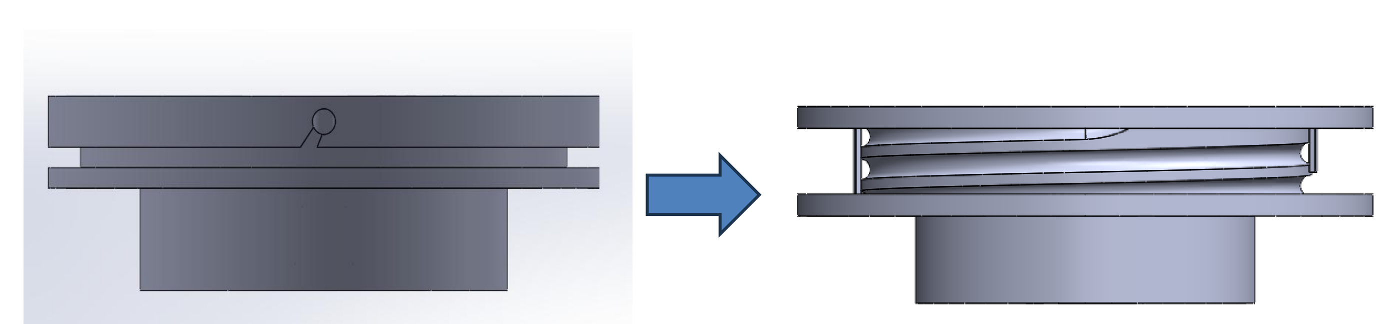
In the new design, the attachment became an individual part, which would be secured after the motors are attached to the instrument. Decreasing the number of bending and breakage.



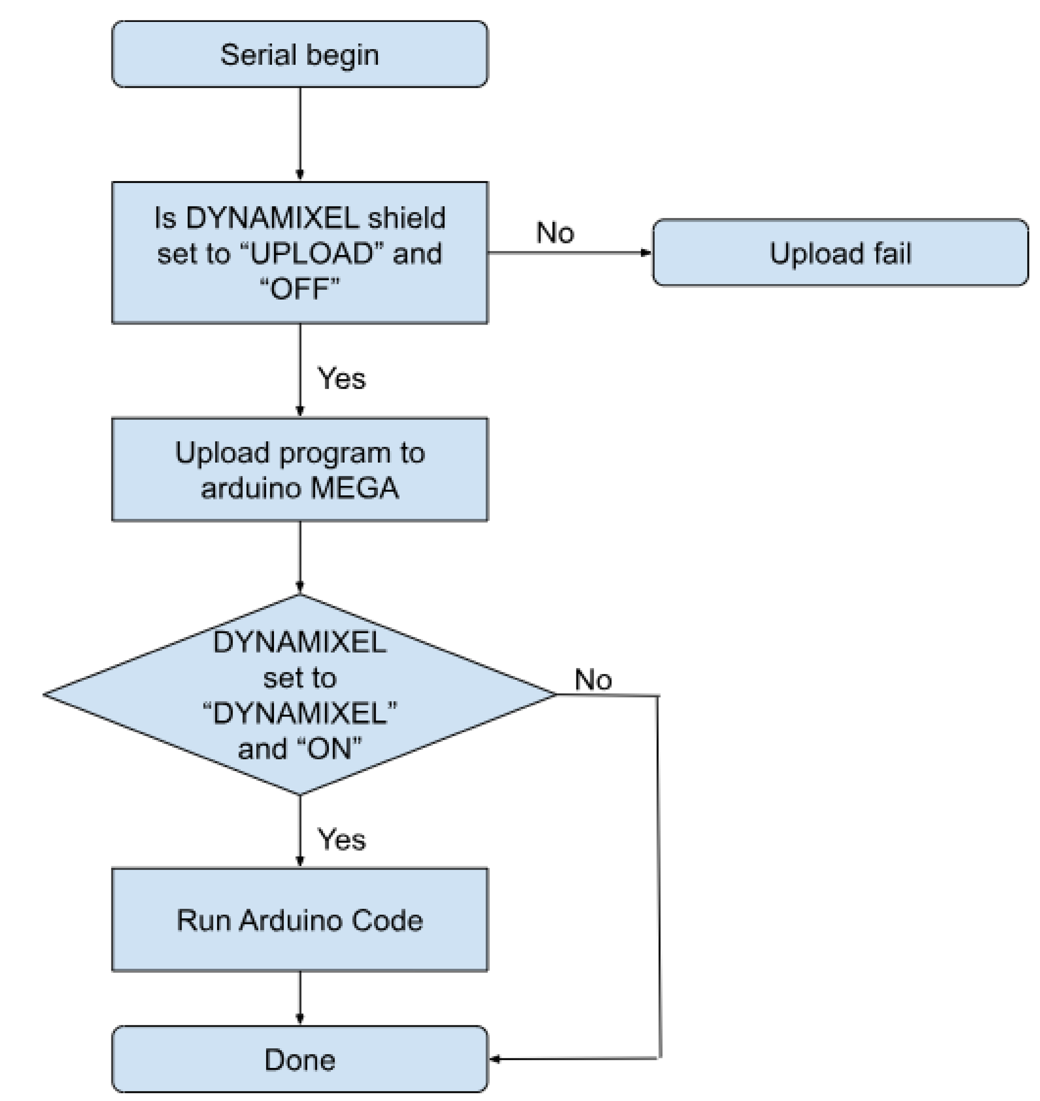
Pulley

In order to decrease the wires slipping and tangling, the redesign of the pulley to a spiral like pulley minimizes both incidents.

The new spiral design also consist of walls that increase the tension between the wires and the pulley, thus decreasing slipping and increasing precision.



Early Software Implementation



Moving Forward

- Implementation of all four motors moving simultaneously
- Learn about ROS (Robot Operating System) in order to visualize the movement of the instrument

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